**University of Koblenz**

**Master's Thesis Proposal**

# Benchmarking Holistic vs Sentence-Level RLAIF for Factual Summarization via Granular Credit Assignment (GCA)

Author: Muhammad Hasnat
Supervisors: Dr. Zeyd Boukhers, Prof. Dr. Frank Hopfgartner
Mentor: Lingxiao Kong
Date: 03 March 2026

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# 1. Introduction

Large language models are increasingly used for multi-sentence generation tasks such as news summarization. In this setting, perceived quality is often dominated by a small number of high-impact factual errors: a wrong entity, a flipped relation, or a fabricated claim can outweigh otherwise fluent writing. In this proposal, reliability refers primarily to factual consistency with the source document, not stylistic polish.

Preference-based fine-tuning, including reinforcement learning from human feedback (RLHF), updates models using comparative judgments rather than token-level supervision. Because collecting high-quality human preferences is expensive, reinforcement learning from AI feedback (RLAIF) replaces humans with an LLM judge, enabling faster iteration but introducing label noise and bias.

This thesis targets one controlled question: for long summaries, are holistic A/B preferences enough to improve factual reliability, or does supervision become more effective when the judge evaluates aligned sentence pairs and those local decisions are aggregated into a summary-level preference? The training objective (DPO), candidate pool, model, and compute budget are held constant; only feedback granularity changes.

## 1.1 Problem Statement

When training summarization models with AI-generated preferences, it is unclear whether coarse, holistic preferences are sufficient for improving factual reliability, or whether they systematically underutilize the available supervision signal in long outputs. The problem is to determine, under controlled conditions, whether sentence-level AI judgments aggregated into summary-level preferences yield more reliable summaries than purely holistic AI judgments.

## 1.2 Research Gap

Prior work shows that fine-grained human feedback can improve preference learning in long-form generation, but evidence is limited for fine-grained AI feedback in realistic, budgeted pipelines where the learning objective remains offline preference optimization. Separately, LLM-as-a-judge studies document prompt sensitivity, position bias, and inconsistency, yet many RLAIF pipelines treat judge labels as ground truth. For long summaries, factual errors are typically sparse and localized, which can create a mismatch between holistic supervision and where the errors actually occur. There is a lack of controlled studies that ,(i) isolate feedback granularity as the only experimental variable, (ii) focus on factual reliability in summarization, and (iii) quantify judge reliability with explicit human auditing and inter-rater agreement.

## 1.3 Proposed Novelty

- A controlled comparison where the only factor that differs is feedback granularity (holistic vs sentence-level), with identical candidate pools, base model, and DPO training recipe.
- A practical Granular Credit Assignment (GCA) aggregation layer that converts sentence-level A/B judgments (with rationales and confidence) into standard summary-level preference pairs usable by DPO, without training a separate reward model.
- A reliability-oriented labeling protocol for LLM judges (evidence-grounded rationales, confidence gating, A/B position randomization, and a human audit) to make reliability claims defensible.
- An error taxonomy and qualitative analysis that links improvements or failures back to specific sentence-level decisions, improving interpretability of preference optimization.

### 1.4 Expected Contributions

This thesis will deliver:

- Empirical evidence on whether sentence-level AI feedback improves factual consistency in news summarization compared to holistic AI feedback under matched compute and the same offline objective (DPO).
- A reproducible pipeline to generate preference datasets at two granularities, including alignment, filtering, and audit artifacts (prompts, thresholds, and logs).
- Quantitative reliability results for an LLM judge on this task, including agreement with humans (Cohen's Kappa and accuracy) and an analysis of failure modes that matter for preference learning. [25]
- Practical guidance on when granular labeling is worth the extra cost (and when alignment/judge noise cancels the gains).

### 1.5 Scope and Assumptions

- Task: single-document news summarization.
- Training: offline preference optimization via Direct Preference Optimization (DPO) with parameter-efficient fine-tuning (QLoRA).
- Judging: one fixed instruction-tuned LLM judge with basic bias mitigations; optional replication with a second judge is used only as a robustness check.
- No retrieval augmentation or external tools are used in generation or evaluation; improvements must come from the supervision signal and optimization.

## 2. Literature Review

### 2.1 Preference-Based Alignment: RLHF, RLAIF, and Offline Optimization

Preference-based alignment collects pairwise comparisons between candidate outputs and converts those judgments into a training signal. This idea comes from preference learning in reinforcement learning, where humans compare short trajectory segments and a reward model is trained from those comparisons [2]. In language generation, this was adapted to summarization by collecting human comparisons, training a reward model, and fine-tuning the policy against that reward [4]. RLHF later popularized this recipe for instruction following at scale [5].

RLAIF replaces human comparisons with judgments produced by a strong language model, reducing cost while introducing new risks. Constitutional AI is a prominent RLAIF formulation that generates critiques and preferences from principles, then uses those preferences for alignment [1]. Follow-up work comparing RLAIF and RLHF reports competitive performance, but emphasizes that outcomes depend on judge prompt design and calibration [6].

Many RLHF pipelines optimize with on-policy methods such as PPO [3]. DPO reframes preference optimization as a closed-form objective, removing explicit reward modeling and on-policy updates while preserving the intent of optimizing pairwise preferences [7]. This thesis adopts DPO so experiments isolate the effect of feedback granularity and judge reliability rather than reinforcement learning dynamics.

## 2.2 Reliability Risks in AI Supervision and LLM-as-a-Judge

Using an LLM as a judge introduces sensitivity to prompt wording, position and verbosity bias, inconsistent scoring across runs, and self-enhancement effects. These issues are documented in benchmark-based studies of LLM-as-a-judge, alongside mitigations such as rubric enforcement, pairwise evaluation, and calibration prompts [10], [11]. A recent survey synthesizes the evidence base on reliability challenges and mitigation strategies [12].

Because this thesis uses AI-generated preferences as training supervision, judge reliability is not only an evaluation concern: it shapes the learning signal. Broader analyses of RLHF and preference optimization highlight failure modes such as label noise and objective mis-specification that can push optimization away from the desired behavior even when training loss improves [8]. Accordingly, the judge is treated as a fallible measurement device and the design includes robustness checks and label-quality controls.

## 2.3 Fine-Grained Feedback and Credit Assignment in Long-Form Generation

Holistic feedback is often too coarse for long outputs because it does not localize which parts caused the preference. The credit assignment problem is central even in classic preference learning, where annotators compare short segments to provide denser supervision [2]. For language generation, fine-grained human feedback yields better reward signals and downstream optimization behavior than purely holistic judgments, especially when errors are localized (for example, a single false sentence in an otherwise good answer) [9].

In factuality evaluation, entailment models trained on sentence-level NLI data underperform on document-level consistency, motivating decomposition into sentence units and aggregation [13]. This thesis adapts that intuition for preference labeling: it collects sentence-pair preferences and aggregates them into a single summary-level preference via GCA. This keeps standard summary-level preference optimization while making supervision more localized and diagnosable.

## 2.4 Factual Consistency in Summarization and Evaluation

News summarization remains a standard benchmark domain, with CNN/DailyMail commonly used for training and evaluation [18], [19]. Despite progress in fluency, factual inconsistency and hallucination persist [20].

Automatic factuality evaluation has progressed along entailment-based and QA-based lines. FactCC detects factual conflicts between the source and the summary [16]. QA-based approaches such as FEQA and QAFactEval compare answers derived from the summary and the source [15], [14]. SummaC shows that sentence-level decomposition and aggregation can improve inconsistency detection [13].

Recent LLM-based evaluation work has moved beyond summary-level Likert scoring. FineSurE provides a fine-grained evaluation framework that assesses summaries at the sentence or key-fact level along multiple dimensions, enabling diagnostic analysis of factuality and coverage trade-offs [23].

## 2.5 Synthesis and Positioning

The literature suggests that preference optimization is practical, but the supervision source and its granularity strongly shape outcomes. RLAIF and LLM-as-a-judge work mostly studies preference quality at the response level [1], [6], [10]–[12]. Fine-grained feedback work shows advantages but often relies on human annotation and may change the optimization signal throughout training [9].

This thesis addresses how to collect and use sentence-level preference labels for long-form summarization while staying compatible with standard offline preference optimization, and while explicitly auditing judge reliability. The novelty is to operationalize GCA as an aggregation layer that turns sentence-level comparisons into conventional summary-level preference pairs for DPO training [7]. Compared with FineSurE, which is primarily an evaluation framework, this method injects localized supervision into the training signal; this should better target the specific sentences where hallucinations occur rather than relying on post-hoc scoring [13], [23].

## 3. Research Questions and Objectives

The overarching objective is to determine whether more granular AI supervision improves factual reliability in news summarization under a controlled, offline preference-learning setup.

### 3.1 Research Questions

- RQ1: Does sentence-level AI feedback aggregated via GCA produce more factually consistent summaries than holistic AI feedback when used for DPO fine-tuning?
- RQ2: How sensitive are the observed effects to alignment strategy (index-based vs semantic alignment) and to judge reliability controls such as confidence gating and A/B order randomization?
- RQ3: What categories of factual errors are most affected by sentence-level supervision (entities, numbers, relations, temporal claims)?
- RQ4: Do gains persist under human auditing and (optionally) a second judge, or are they judge-specific artifacts?

### 3.2 Hypotheses

- H1: Sentence-level aggregated preferences reduce factual inconsistency more than holistic preferences because they localize the supervision signal on long outputs.
- H2: Better alignment and reliability controls reduce label noise and training variance; if alignment noise is too high, it can mask the benefits of GCA.
- H3: Improvements are largest for localized errors (entity and relation mistakes) rather than global attributes such as style.
- H4: If improvements reflect genuine factuality gains, they remain visible under at least one independent evaluator (automatic metrics and/or human audit).

## 4. Methodology

This thesis follows a controlled experimental design to test whether feedback granularity in AI-supervised preference learning affects factual reliability in news summarization. The base model, candidate pool, dataset split, training objective (DPO), and compute budget are held constant across conditions. The only manipulated factor in the main experiment is the granularity of the preference labels used for optimization (holistic versus sentence-level labels aggregated into a summary-level preference).

### 4.1 Study Design

The study compares two preference-labeling regimes under the same offline preference optimization objective. The holistic regime collects one A/B preference per summary pair. The sentence-level regime

collects per-sentence preferences and aggregates them into a single summary-level preference using a transparent rule (GCA). Judge reliability is treated as a measurement problem: labels are filtered and audited using consistency checks, confidence gating, and a human spot check to support defensible claims.

### 4.2 Sentence Alignment as a Noise Source (and Ablation)

Sentence alignment is treated as a first-class source of noise. Summaries often involve many-to-one mappings and significant rephrasing, so alignment errors can contaminate sentence-level preferences and hide true effects. The main sentence-level pipeline uses greedy semantic-similarity matching with a minimum similarity threshold, plus explicit handling of unmatched sentences. An ablation compares this semantic alignment to a simpler index-based alignment to quantify how alignment quality affects downstream label noise and model performance.

### 4.3 Granular Credit Assignment via Aggregation (GCA)

GCA is an aggregation layer between sentence-level judgments and the final preference pairs used by DPO. For each candidate pair, sentence-level wins are counted, ties are allowed, and the final summary preference is derived by a simple majority rule. Majority aggregation is chosen for transparency and to avoid introducing an additional learned component, keeping the main comparison focused on supervision granularity. To reduce noise, pairs are discarded when the judge expresses low confidence or when the aggregated outcome is ambiguous. A secondary ablation tests a confidence-weighted variant to assess whether self-reported confidence improves label quality.

### 4.4 Primary vs Secondary Experiments (MoDPO as Optional Extension)

The primary goal is to produce clear empirical evidence for the effect of GCA-driven granularity (holistic vs sentence-level) on factual reliability. Multi-objective trade-offs can be explored, but only after the primary comparison is established.

As an optional extension, the study may apply Multi-Objective DPO (MoDPO) by tagging sentence-level comparisons with objective labels (e.g., faithfulness vs coverage) and conditioning optimization on the objective indicator [24]. This extension is explicitly separated from the main experiment to avoid diluting the core evidence about GCA.

### 4.5 Judge Reliability Plan and Human Audit

To make results defensible, judge reliability is measured and controlled rather than assumed.

- Order randomization: swap A/B positions; if preferences disagree, mark as unstable and filter or send to audit.
- Confidence gating: keep only labels above a conservative threshold emitted by the judge; report coverage (kept vs dropped).
- Evidence requirement: rationales must cite the specific claim, entity, or sentence that determines the decision; non-evidential outputs are discarded.
- Prompt robustness: run two paraphrased prompts on a subset and measure preference stability; optionally filter inconsistent cases.

Human audit (planned sample size):

- Sample: 100 summary pairs, stratified to include (i) high-confidence agreements, (ii) borderline low-confidence cases, and (iii) cases where holistic and sentence-level pipelines disagree.
- Annotators: 2 human raters (trained with a short rubric) label the same pairs for factual consistency with the source. If disagreement is high, add a third adjudicator for the final label.
- Metrics: Cohen's Kappa for inter-rater reliability; judge-vs-human agreement reported as accuracy and Cohen's Kappa against the majority human label. [25]
- Outcome: report common failure modes (e.g., missed entity swaps, over-weighting fluency, ignoring omissions) and how filtering affects downstream training.

## 5. Ethical and Validity Considerations

### 5.1 Ethical and Practical Considerations

The experiments use publicly available news articles and summaries. No personally sensitive data is intentionally collected. Because model outputs may contain inaccuracies, the thesis reports factuality-focused evaluation, includes qualitative error analysis, and avoids presenting generated content as ground truth. Compute usage is managed through dataset subsetting and parameter-efficient fine-tuning to reduce environmental and resource costs.

### 5.2 Threats to Validity and Mitigations

- Judge bias/instability: mitigated via order randomization, confidence gating, prompt robustness checks, and human auditing.
- Alignment noise: measured via index vs semantic alignment ablation; ambiguous or poorly aligned pairs can be filtered.
- Metric limitations: use complementary factuality metrics (NLI-based and QA-based) and validate trends with human audit.
- Overfitting to judge artifacts: evaluate with at least one independent evaluator and report qualitative examples.

## 6. Implementation

Figure 1 summarizes the pipeline with inputs and outputs for each step. Candidate summaries are generated once from a fixed base policy and cached to form a fixed candidate pool. Preference labels are produced in two regimes: (A) holistic A/B judgments over full summaries, and (B) sentence-level judgments over aligned sentence pairs followed by GCA aggregation into summary-level preference pairs. The resulting preference datasets are used for DPO fine-tuning with QLoRA. Evaluation then measures factual consistency, reports judge reliability, and performs a human audit and qualitative error analysis.

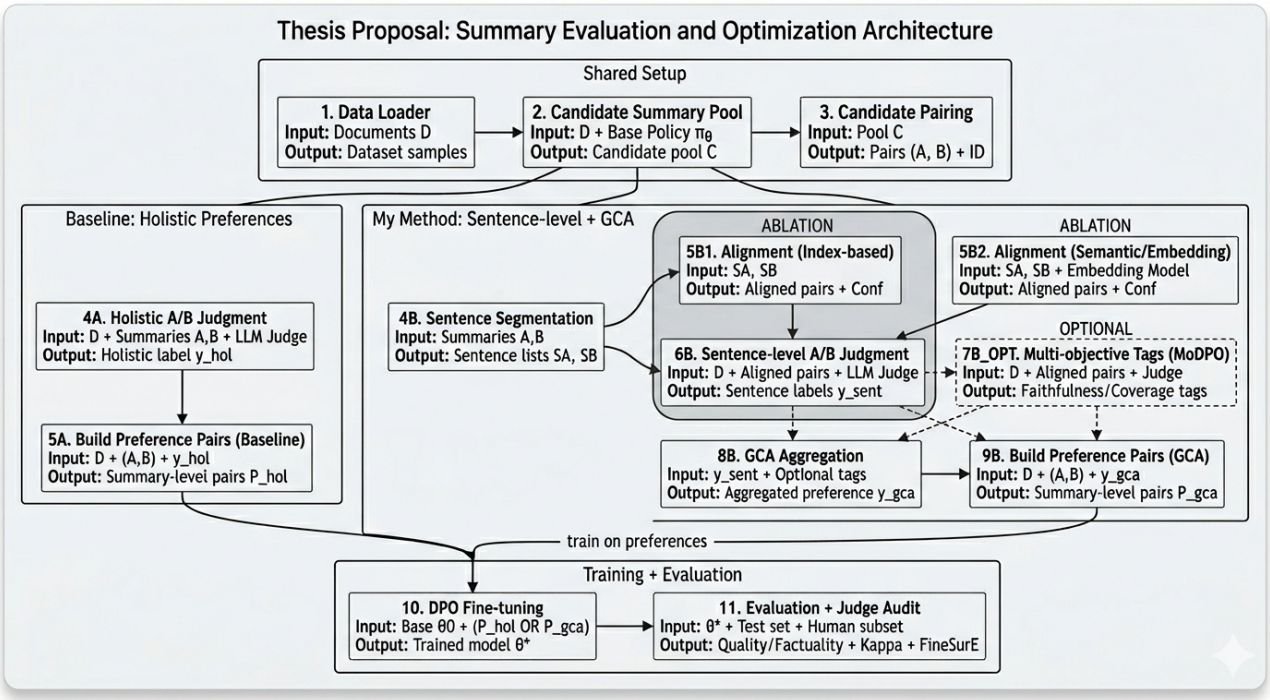


Figure 1. training and evaluation pipeline with explicit inputs and outputs.

Task and dataset: single-document abstractive summarization in the news domain using a compute-controlled subset of CNN/DailyMail. Inputs are truncated to a fixed maximum length and output length is constrained by decoding parameters for fair comparisons across regimes. A train/validation/test split is created with a fixed random seed.

Models and training: the policy model is a single instruction-tuned model in the 7B parameter class. Fine-tuning uses QLoRA and DPO [7]. All training hyperparameters are held constant across regimes.

Labeling regimes: both regimes use a fixed LLM judge prompt requesting (i) a discrete A/B preference, (ii) an evidence-grounded rationale, and (iii) a confidence indicator. For sentence-level labeling, sentences are segmented consistently and aligned using semantic matching; an index-based alignment ablation is retained.

Evaluation: factual consistency is the primary target. Automatic metrics include NLI-based and QA-based factuality scorers (e.g., SummaC and QAFactEval) and standard similarity metrics (ROUGE, BERTScore). FineSurE is used as a diagnostic evaluator (analysis support), not as the core optimization target [23]. Statistical testing uses paired bootstrap resampling over the test set (e.g., 10,000 resamples), reporting 95% confidence intervals and two-sided significance at $p < 0.05$, alongside effect sizes (mean and median deltas). A qualitative error analysis links gains or failures back to sentence-level decisions.

## 7. Timeline

Critical risks (judge reliability, alignment noise, and evaluation) are addressed early. If full-scale fine-tuning is unstable or compute-constrained, the fallback is to reduce the preference dataset size while keeping the same controlled design and reporting learning curves with uncertainty.

| Phase | Activities | Deliverables |
|---|---|---|
| 1: Setup and baselines | Finalize dataset subset and preprocessing; implement candidate generation; implement evaluation scripts; reproduce base-model results. | Reproducible data pipeline; baseline evaluation report. |
| 2: Judge prompts and audits | Design judge prompts for holistic and sentence-level labeling; implement reliability controls; run human audit. | Validated labeling protocol; documented judge failure modes. |
| 3: Preference dataset construction | Generate holistic preference pairs; generate sentence-level judgments; aggregate with GCA; apply filtering and quality checks. | Two cleaned preference datasets with metadata and filters. |
| 4: DPO fine-tuning | Run two DPO fine-tunes under identical configurations; track compute and training curves; select checkpoints via validation. | Two tuned models and training logs. |
| 5: Evaluation and analysis | Run full test evaluation; uncertainty estimation; qualitative error analysis; judge vs human agreement analysis. | Results draft; error taxonomy; discussion draft. |
| 6: Writing and revision | Draft thesis chapters; iterate with supervisors; finalize artifacts and appendix (prompts, scripts). | Final thesis submission package. |

## 8. Expected Outcomes and Analysis Plan

- Reliability outcome: whether the sentence-level regime consistently improves factuality metrics and human-audited factual correctness relative to the holistic regime. A meaningful improvement is defined as a consistent gain on at least two factuality measures (automatic and/or human audit) without a statistically significant degradation in the other.
- Stability outcome: whether the sentence-level regime reduces judge noise and training variance once reliability controls are applied (and whether alignment noise dominates).
- Interpretability outcome: whether sentence-level rationales make it easier to diagnose which errors were corrected and which persist after fine-tuning.

If sentence-level supervision does not outperform holistic supervision, the thesis will report a rigorous negative result and explain it through measured alignment noise, aggregation failures, or judge unreliability. In either case, the study will produce actionable guidance on when granular labeling is worth the additional labeling cost.

## 9. Conclusion

This proposal presents a controlled study of feedback granularity in AI-supervised preference learning for news summarization. By holding the model, candidate outputs, and training recipe constant and varying only the granularity of AI judgments, the thesis aims to produce clear evidence about whether granular credit assignment improves factual reliability. The work contributes empirical findings and a reproducible methodology for reliability-aware AI feedback pipelines.

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